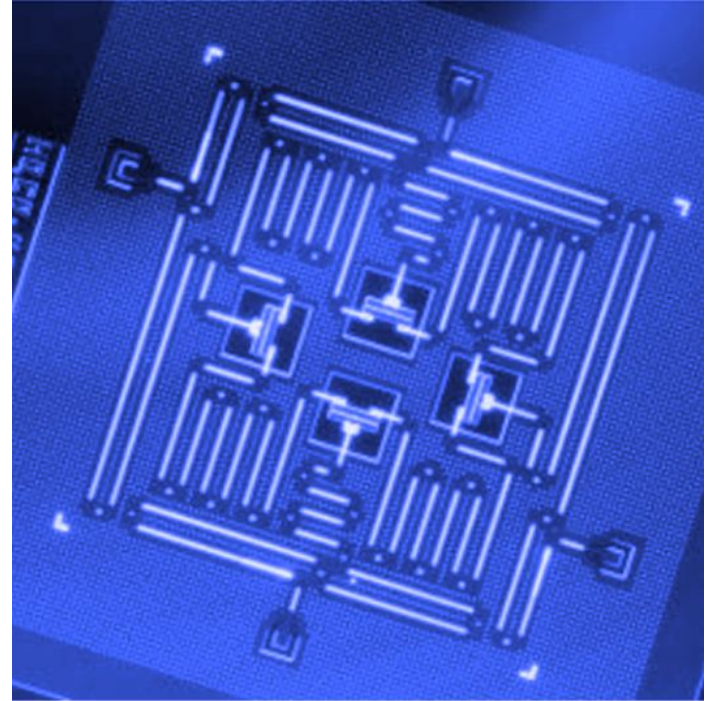


Design and simulation of superconducting qubits

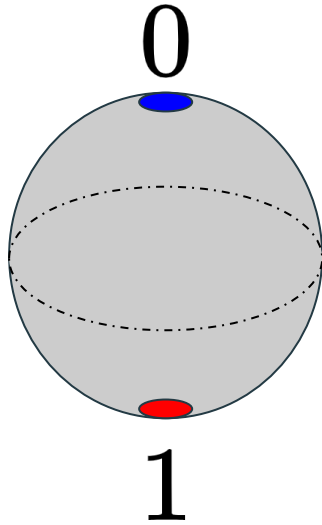
Danilo Labranca



By Jay M. Gambetta, Jerry M. Chow & Matthias Steffen -
<https://www.nature.com/articles/s41534-016-0004-0>, CC BY 4.0,
<https://commons.wikimedia.org/w/index.php?curid=61462110>

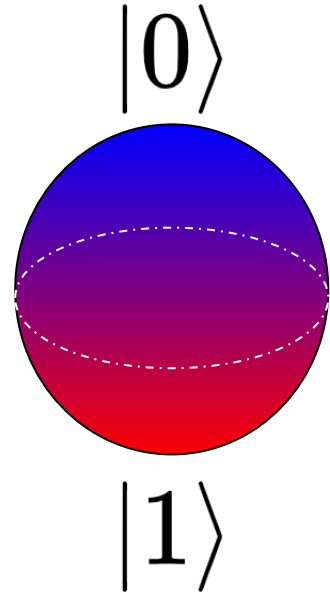
What is a qubit?

- Qubits are two state quantum systems
- They are the quantum mechanical analogous of the bit

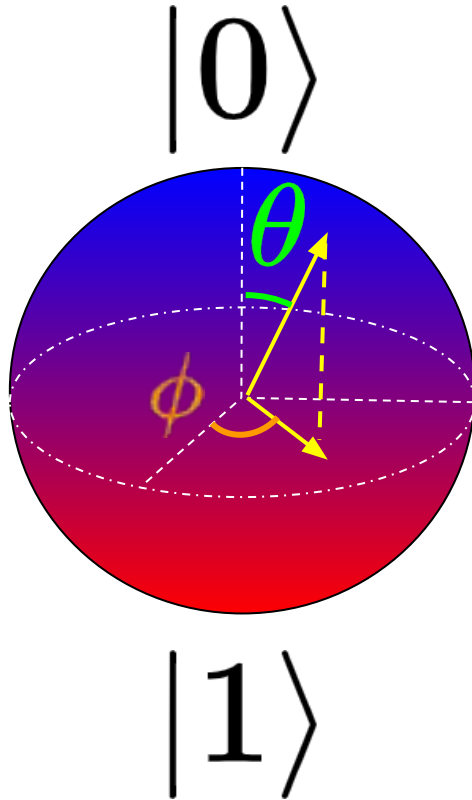


**Classical
Bit**

**Quantum
Bit**



What is a qubit?

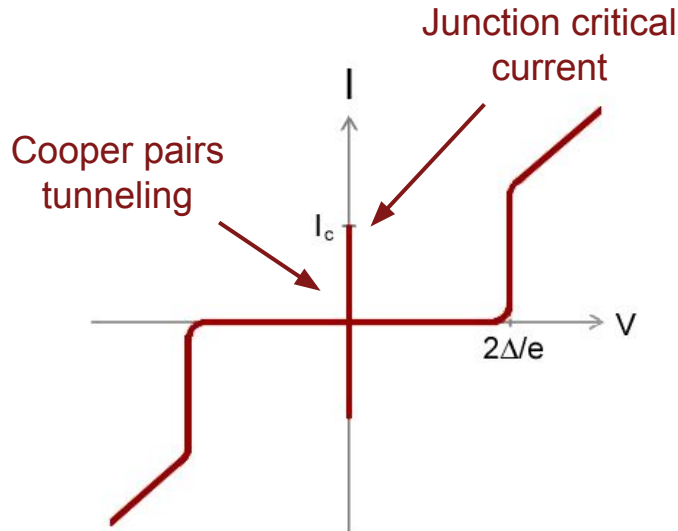
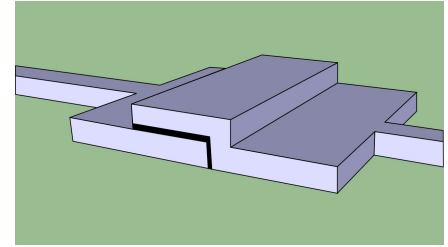
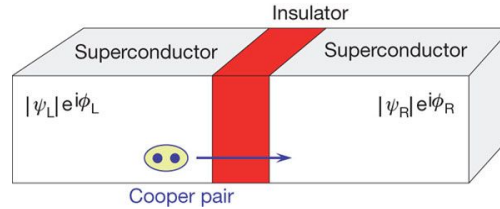


Qubits can store a lot more information than a classical bit

$$\begin{aligned} |\psi\rangle &= \alpha|0\rangle + \beta|1\rangle \\ &= \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \end{aligned}$$

Superconducting qubits

Superconducting qubits are based on Josephson Junctions or STJ (Superconducting Tunnel Junctions)



Cooper pairs in the SC can go through the insulator barrier thanks to quantum tunneling. Thus, a current can be observed with zero voltage drop across the junction.

It can be proved that the junction is equivalent to a non-linear inductor with inductance:

$$L(\Phi) = \frac{\Phi_0}{2\pi I_C} \frac{1}{\cos(2\pi\Phi/\Phi_0)}$$

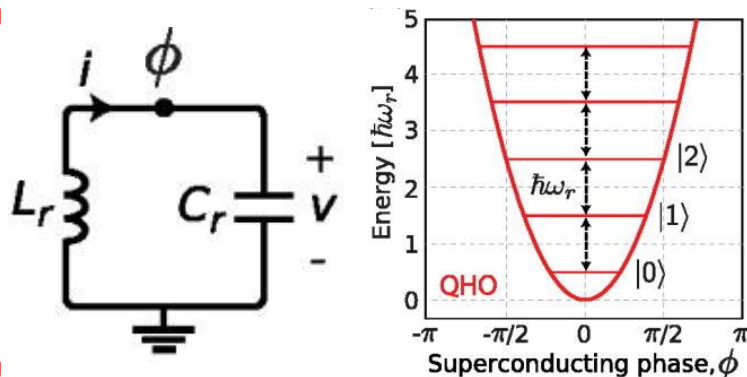
L_J

Superconducting qubits are essentially LC oscillators with Josephson Junctions replacing the inductance.

Harmonic oscillator

$$E_C = \frac{e^2}{2C_r} \quad E_L = \left(\frac{\Phi_0}{2\pi}\right)^2 \frac{1}{L_r}$$

$$H = 4E_C n^2 + \frac{1}{2} E_L \phi^2$$



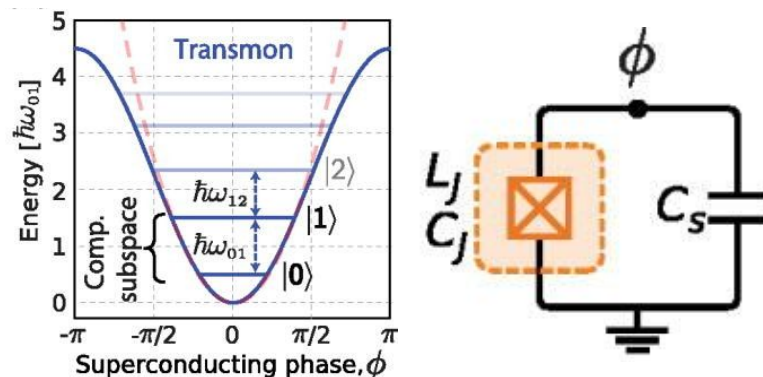
$$\omega_{01} = \omega_{12} = \omega_r$$

$$n = \frac{Q}{2e} \quad \Phi_0 = \frac{h}{2e} \quad \phi = \frac{2\pi\Phi}{\Phi_0}$$

Qubit

$$E_C = \frac{e^2}{2(C_J + C_S)} \quad E_J = L_J I_C^2$$

$$H = 4E_C n^2 - E_J \cos \phi$$

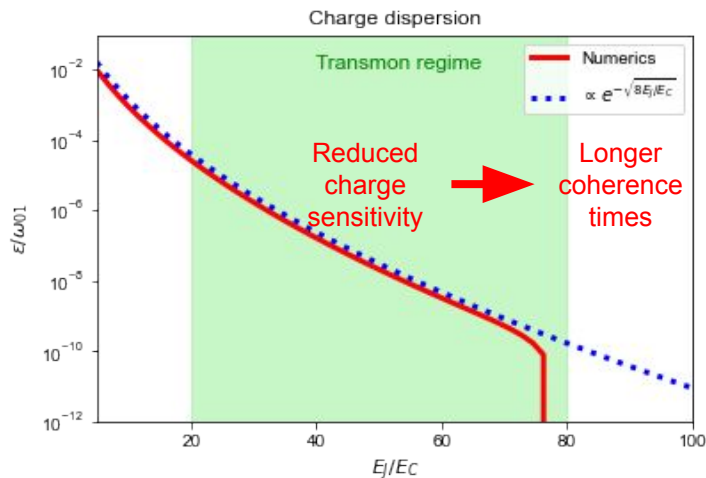


$$\omega_{01} \neq \omega_{12}$$

The transmon qubit

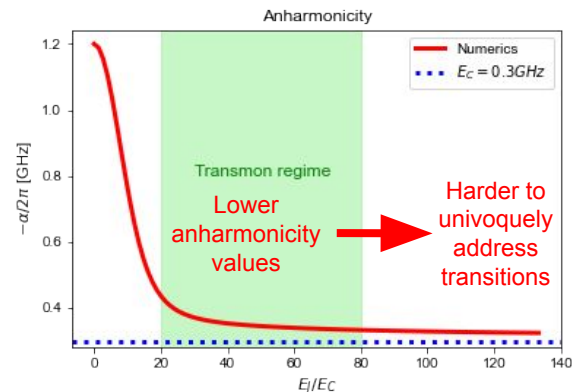
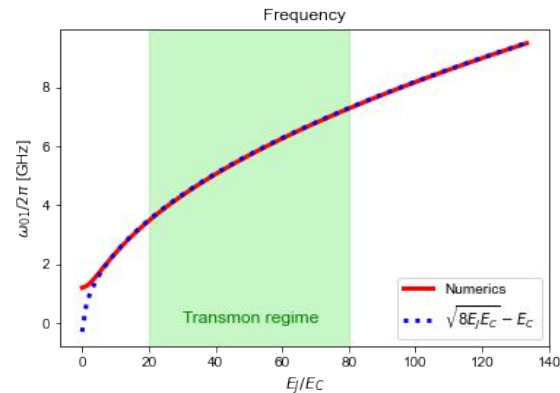
The most used and studied superconducting qubit is the transmon.

It is a charge qubit operated in a specific regime in which: $E_J/E_C \gg 1$



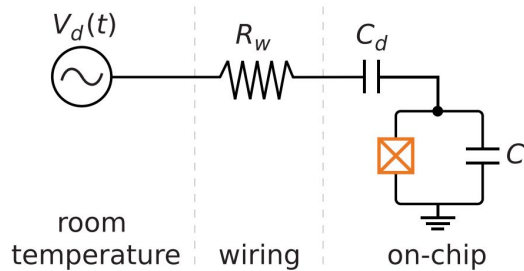
$$E_C = \frac{e^2}{2(C_J + C_S)} \quad E_J = L_J I_C^2$$

$$H = 4E_C n^2 - E_J \cos \phi$$



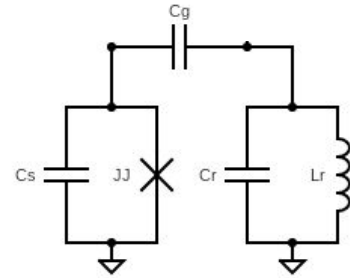
Circuit Quantum Electrodynamics (cQED)

Qubit control

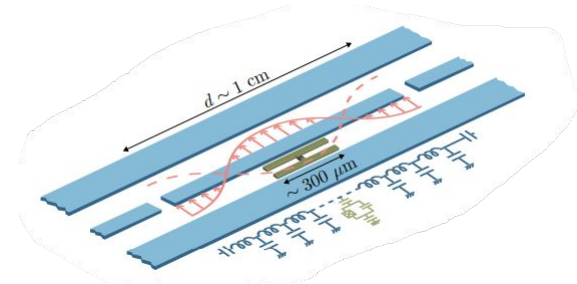
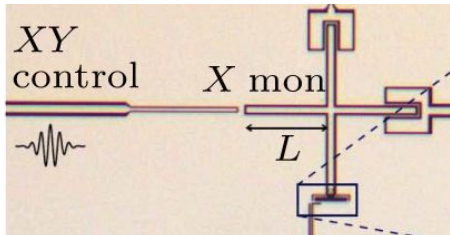


cQED is the study of the interaction between non linear superconducting circuits (e.g. qubits) with electromagnetic fields

Qubit Readout



Qubit Resonator

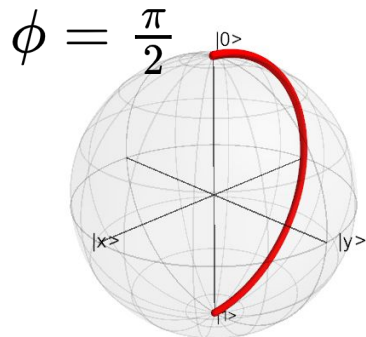
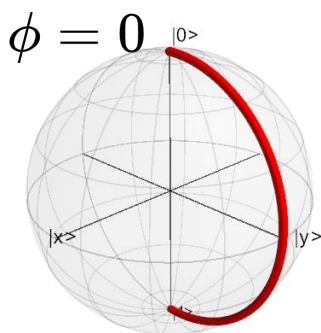


cQED - Qubit control

Superconducting qubits can be controlled using capacitively coupled drive-lines

$$H_d = \overbrace{\left[\frac{\hbar\omega_q}{2} \sigma_z \right]}^{\text{Qubit}} - \overbrace{\left[i\Omega V_d(t)(\sigma_+ - \sigma_-) \right]}^{\text{Drive Interaction}}$$

in the frame rotating at $\omega_q \rightarrow \tilde{H}_d \approx -\frac{\Omega}{2} V_0(t)(I\sigma_x + Q\sigma_y) \quad \omega_d = \omega_q$



$$I = \cos(\phi)$$

$$Q = \sin(\phi)$$

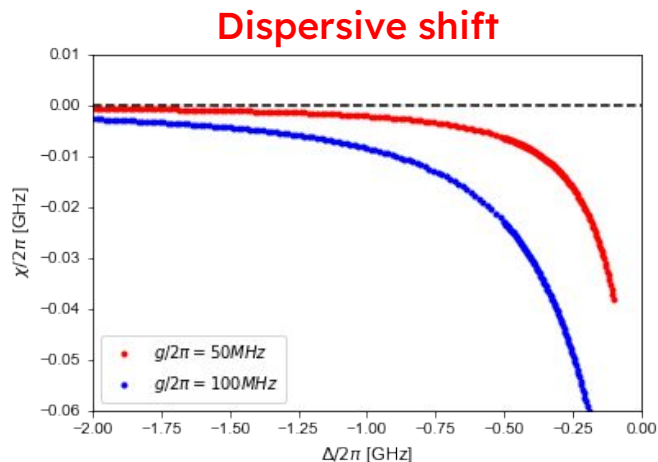
By choosing the phase of the signal, we choose the rotation axis on the Bloch sphere

We can prepare the qubit in whatever state we want via a series of x and y rotations

cQED - Qubit readout

Jaynes-Cummings
Hamiltonian well describes
the interaction between
superconducting resonators
and qubits

$$H_{JC} = \underbrace{\hbar\omega_r a^\dagger a}_{\text{Resonator}} + \underbrace{\frac{\hbar\omega_q}{2} \sigma_z}_{\text{Qubit}} + \underbrace{\hbar g(a^\dagger \sigma_- + a \sigma_+)}_{\text{Interaction}}$$



Dispersive regime:

$$\Delta = \omega_q - \omega_r$$

1

$$\Delta \neq 0$$

2

$$|g/\Delta| \gg 1$$

$$H_{JC} \approx \hbar(\underbrace{\omega_r + \chi\sigma_z})a^\dagger a + \frac{\hbar\omega_q}{2}\sigma_z, \text{ with } \chi \approx \frac{g^2}{\Delta}$$

**Resonator mode frequency is
qubit-state dependent**

Superconducting qubits as quantum sensors

The Qub-IT experiment wants to build a single photon detector based on superconducting transmon qubits

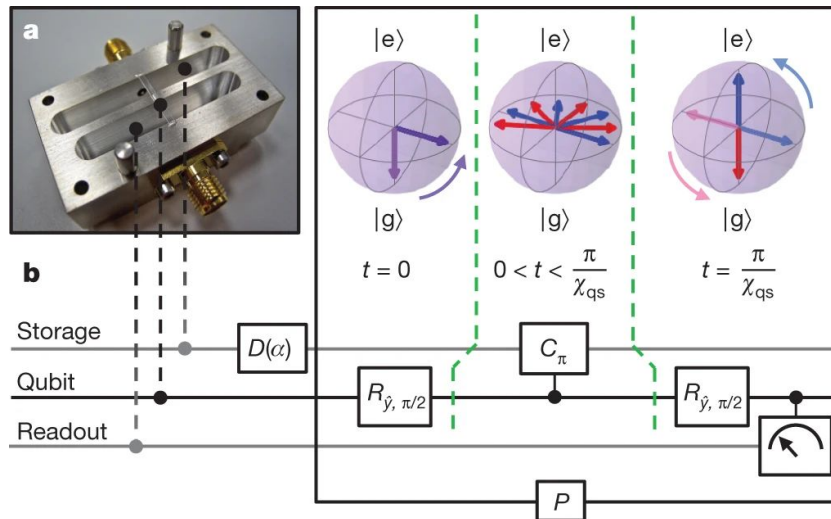
The main idea is to couple the transmon with the photon (via storage resonant cavity or a transmission line)

$$\omega_q \neq \omega_{\text{photon}}$$

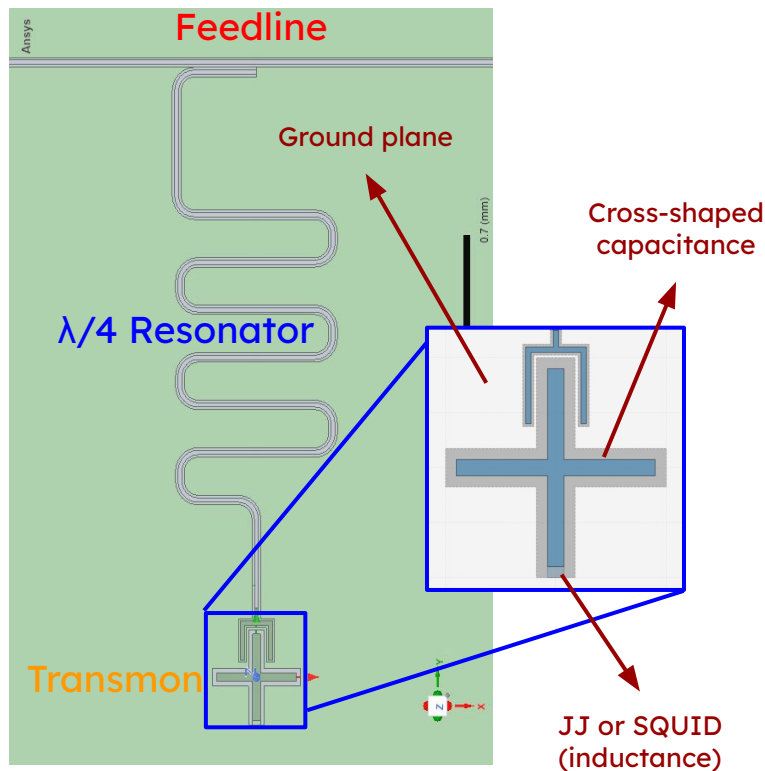


Avoiding the absorption, the photon causes a phase shift in the qubit state

Controlling the qubit with a Ramsey sequence, the presence (or the absence) of a photon can be mapped in the state of the qubit

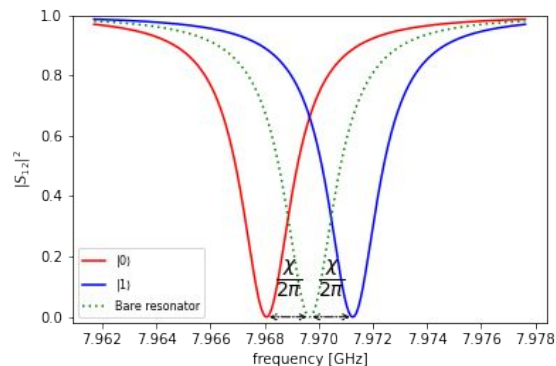


Designing superconducting qubits



- The design of superconducting qubits involves the definition of the chip geometries.
- Usually the **transmon** is capacitively coupled to a **resonator** ($\lambda/4$ or $\lambda/2$), which will be used for the qubit readout.
- The resonator can be coupled to a waveguide (the **feedline**) itself. In this way the qubit state is defined by testing the feedline transmission.

Feedline transmission as a function of frequency



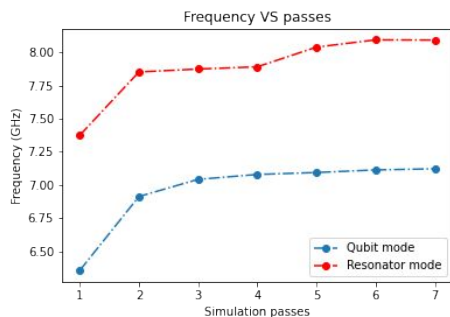
Simulation of superconducting qubits

- Simulations performed with the Energy Participation Ratio method (EPR)
- Ansys EM → calculates the linear electromagnetic oscillation modes of the circuit
- Calculating also the amount of energy stored in the Josephson Junction (participation ratio), the EPR enables to extract the main parameters of the circuit (frequencies, anharmonicity and dispersive shifts)

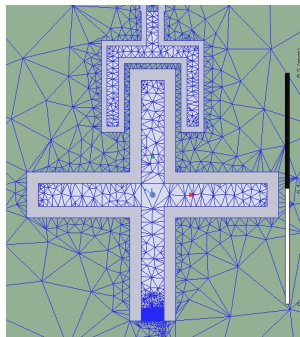
Participation
ratio



$$p_m := \frac{\text{Inductive energy stored in the junction}}{\text{Total inductive energy of mode } m}$$



Ansys simulations
results and mesh
for qubit oscillating
mode



$$\chi_{qr} = p_q p_r \frac{\hbar \omega_q \omega_r}{4E_J}$$

$$\alpha_m = p_m^2 \frac{\hbar \omega_m^2}{8E_J}$$

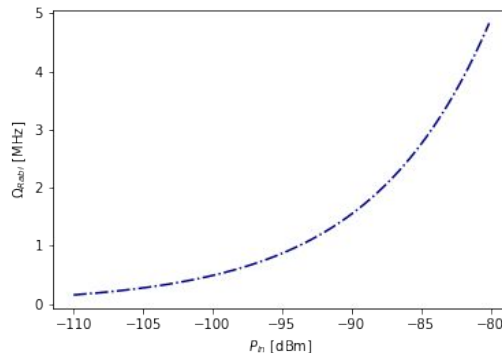
First qubit - Results

$$E_J/E_C \approx 63$$

$$T_1 \approx 10.32 \mu s$$

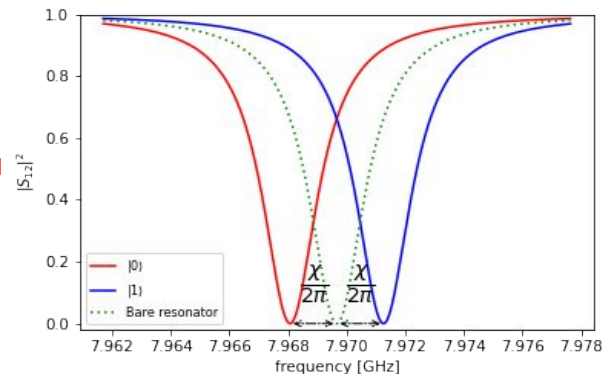
	EPR results	Expected values*
$\omega_{01}/2\pi$	6.81 GHz	6.97 GHz
$\omega_f/2\pi$	8.09 GHz	8.04 GHz
$\alpha/2\pi$	-343 MHz	-325 MHz
$\chi/2\pi$	-1.38 MHz	-1.69 MHz

*: based on equivalent circuit calculations based on capacities and inductances extracted from Q3D simulations



Qubit Rabi frequency as a function of drive power

Feedline transmission as a function of frequency

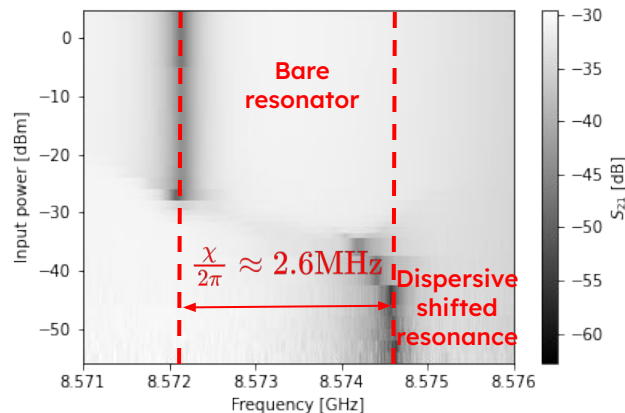
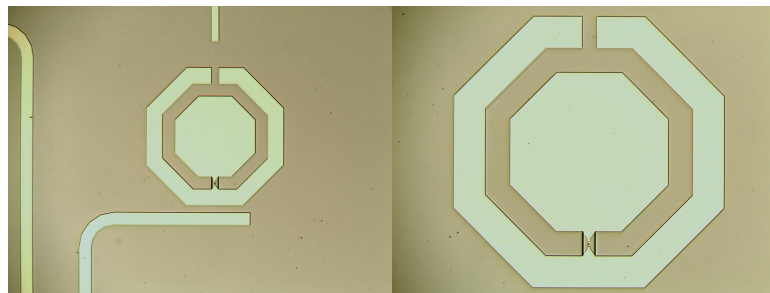


Next steps

- Improve qubit T_1 values to reach a target closer to 100 μs by further detuning the qubit and resonator
- Converge on a definitive design and produce the first chip
- Characterize the chip and compare the results from simulation with the experimental measurements (frequencies, dispersive shifts and couplings, anharmonicity)
- Design and simulate a transmon in a 3D cavity (longer coherence times)
- Use qubit coupled to a transmission line as an ultra-sensitive microwave power meter (collaboration with NIST - Boulder)

Qubit measurement in cryogenic lab

- First measurement of a superconducting qubit produced by NIST
- Measurement of the qubit-resonator dispersive shift
- At high power the dispersive interaction is cancelled and the resonator behaves as the qubit does not exist
- At low power the interaction kicks in and the resonator frequency is shifted by $\chi/2\pi$
- Next steps:
 - Address qubit transition frequency with two-tone spectroscopy
 - Define qubit control pulses
 - Characterize qubit relaxation and dephasing time



Other activities



DART
WARS

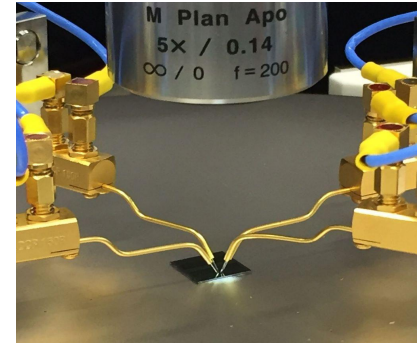
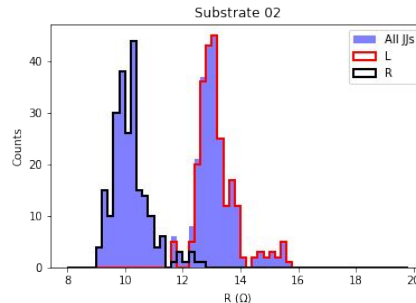
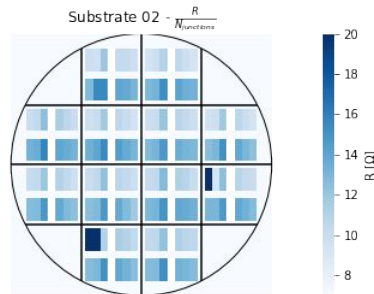
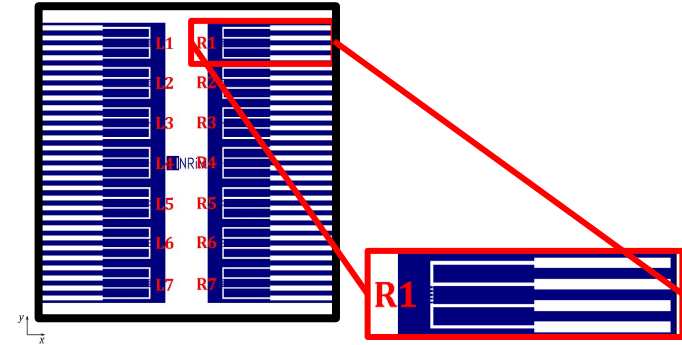


INRiM
ISTITUTO NAZIONALE
DI RICERCA METROLOGICA



Istituto Nazionale di Fisica Nucleare

- Characterization of room temperature normal resistance of Josephson-Junctions produced at Inrim
- 4-terminals measurements are performed with a probe-station and a parameter analyzer
- The goal is to evaluate the dispersion in the resistance value of the single JJs



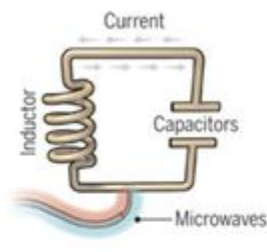
PhD 1st year activities

- PhD Schools
 - INFN school of statistics 2022 - May 15th - 20th 2022
- Conferences
 - 15th Pisa Meeting on Advanced Detectors - May 22nd - 28th 2022
- Exams
 - Python programming - 1 CFU
 - Introduction to Deep Learning for Physicist - 2 CFU
- Multidisciplinary exams:
 - Cultural aspects of the climate crisis - 1 CFU

Backup slides

How to build a qubit?

- There are various physical implementations for qubits



Superconducting loops

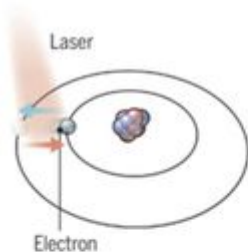
A resistance-free current oscillates back and forth around a circuit loop. An injected microwave signal excites the current into superposition states.

Longevity (seconds)

0.00005

Logic success rate

99.4%



Trapped ions

Electrically charged atoms, or ions, have quantum energies that depend on the location of electrons. Tuned lasers cool and trap the ions, and put them in superposition states.

>1000

99.9%

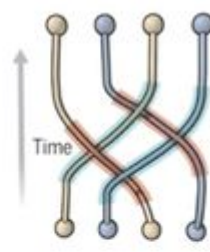


Silicon quantum dots

These "artificial atoms" are made by adding an electron to a small piece of pure silicon. Microwaves control the electron's quantum state.

0.03

~99%

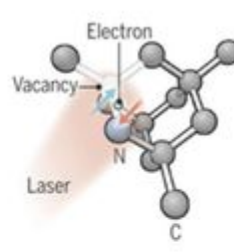


Topological qubits

Quasiparticles can be seen in the behavior of electrons channeled through semiconductor structures. Their braided paths can encode quantum information.

N/A

N/A



Diamond vacancies

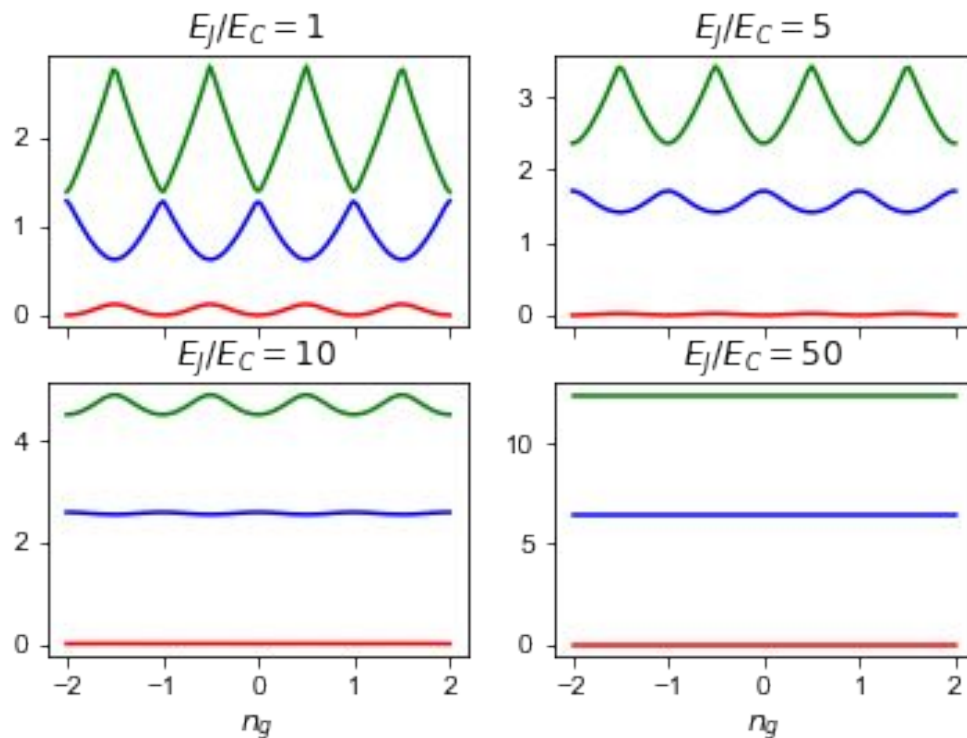
A nitrogen atom and a vacancy add an electron to a diamond lattice. Its quantum spin state, along with those of nearby carbon nuclei, can be controlled with light.

10

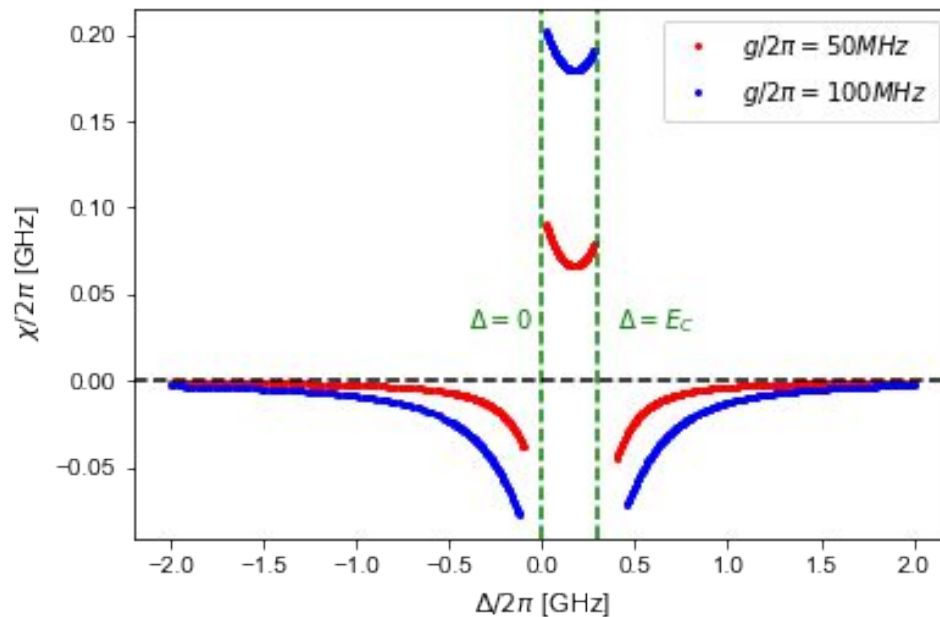
99.2%

<https://www.forbes.com/sites/moorinsights/2019/09/16/quantum-computer-battle-royale-upstart-ions-versus-old-guard-superconductors/?sh=27f049702cb8>

Transmon - Charge dispersion

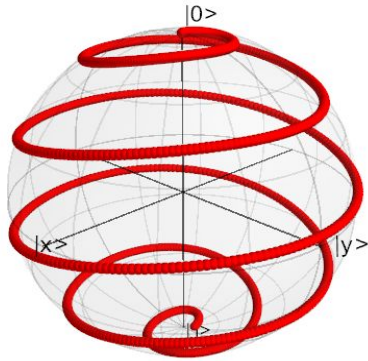


Dispersive regime

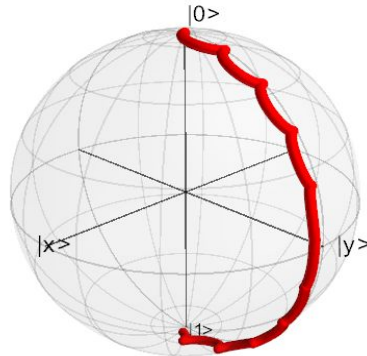


cQED - Qubit control

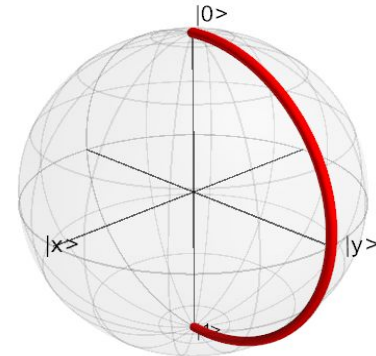
LAB frame



Qubit frame



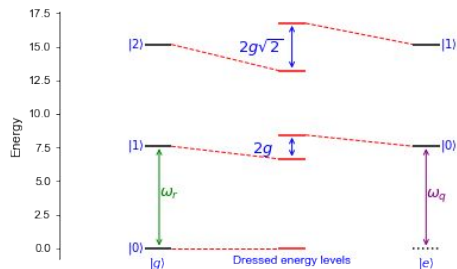
Qubit frame RWA



cQED - Qubit readout

Resonant regime

$$\Delta = 0$$

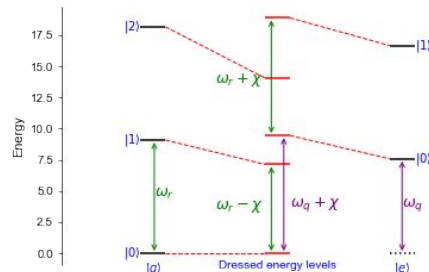


Maximally entangled qubit-resonator state. Qubit is not in a well-defined state.

$$\Delta = \omega_q - \omega_r$$

Dispersive regime

$$\Delta \neq 0 \quad |g/\Delta| \gg 1$$

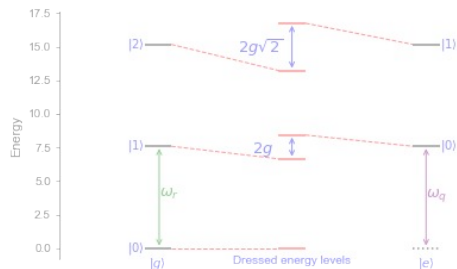


Detuning avoids direct energy exchange between qubit and resonator.
Resonator frequency is qubit-state dependent.

cQED - Qubit readout

Resonant regime

$$\Delta = 0$$



Maximally entangled qubit-resonator state. Qubit is not in a well-defined state.

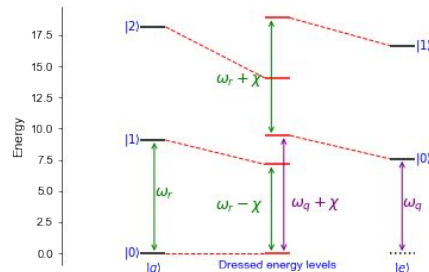
$$\Delta = \omega_q - \omega_r$$

Dispersive interaction term

$$\hbar g(a^\dagger \sigma_- + a \sigma_+) \approx \hbar \chi a^\dagger a \sigma_z$$

Dispersive regime

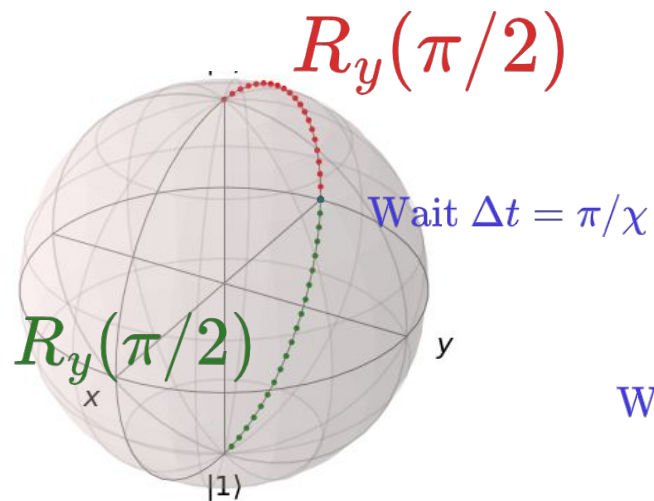
$$\Delta \neq 0 \quad |g/\Delta| \gg 1$$



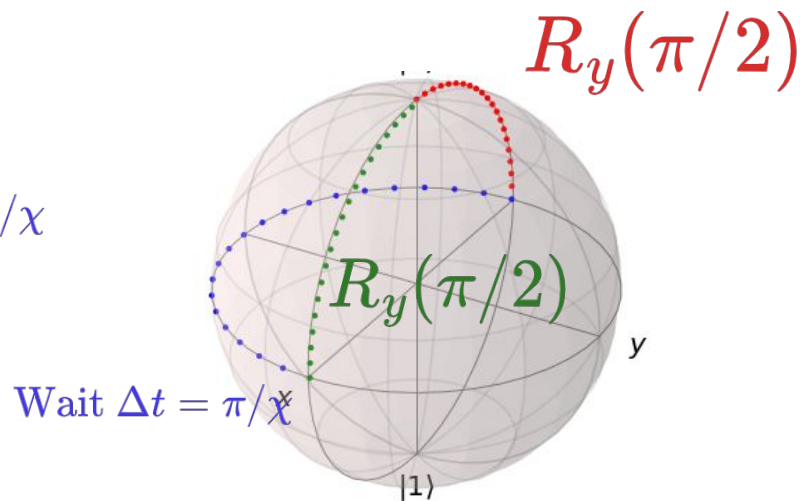
Detuning avoids direct energy exchange between qubit and resonator.

Qubit as a photon detector

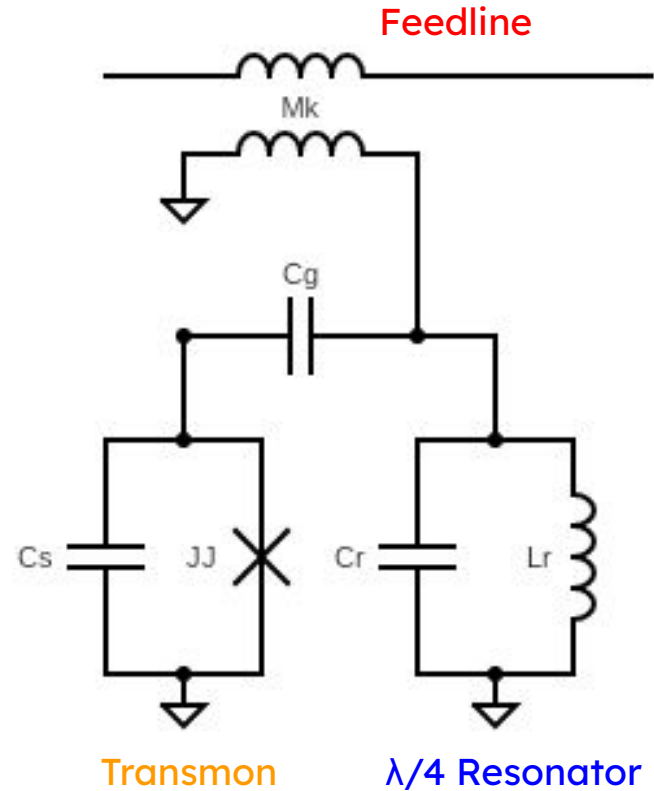
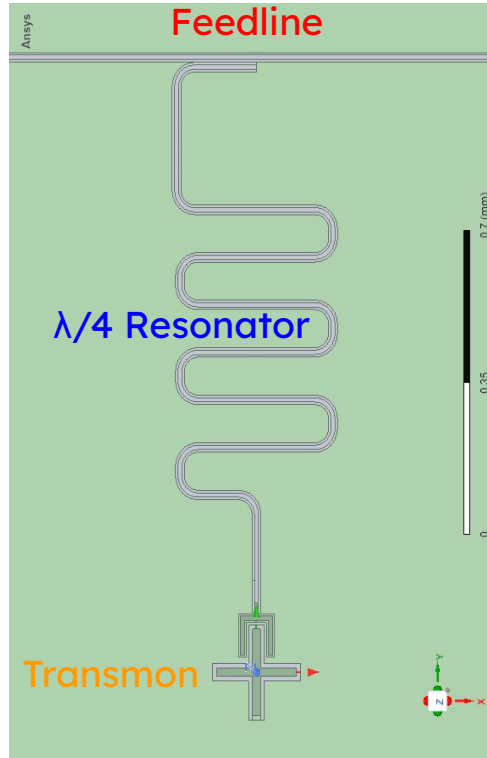
Ramsey - 0 photon



Ramsey - 1 photon



Design equivalent circuit



Other designs



Other designs

